

Docker-01

1. Setup a virtual machine using Ubuntu image.

- Create a new Ec2 instance and new name and AMI Ubuntu

☰ [EC2](#) > [Instances](#) > Launch an instance

Name and tags [Info](#)

Name

[Add additional tags](#)

▼ **Application and OS Images (Amazon Machine Image)** [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

[Recents](#) | [My AMIs](#) | [Quick Start](#)

 Amazon Linux	 macOS	 Ubuntu	 Windows	 Red Hat	 SUSE Linux	 Debian
---	--	---	--	--	---	---

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

- Select family size t3 small and keypair

▼ **Instance type** [Info](#) | [Get advice](#)

Instance type

t3.small Free tier eligible

Family: t3 2 vCPU 2 GiB Memory Current generation: true

On-Demand RHEL base pricing: 0.0496 USD per Hour

On-Demand Windows base pricing: 0.0392 USD per Hour

On-Demand SUSE base pricing: 0.0518 USD per Hour On-Demand Linux base pricing: 0.0208 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0243 USD per Hour

[Additional costs apply for AMIs with pre-installed software](#)

☐ All generations [Compare instance types](#)

▼ **Key pair (login)** [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the i

Key pair name - required

[Create new key pair](#)

- Select Networking settings

▼ **Network settings** [Info](#)

Network [Info](#)

vpc-096ec8a65d98c6729

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

[Compare security group rules](#)

[X](#)

VPC: vpc-096ec8a65d98c6729

Security groups that you add or remove here will be added to or removed from all your network interfaces.

➤ Select storage 20Gb

▼ Configure storage [Info](#)

1x GiB Root volume, 3000 IOPS, Not encrypted

[Add new volume](#)

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance volumes from the AMI will be accessible from the instance

➤ Now connect with Pem key

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ ssh -i "aws.pem" ubuntu@ec2-3-87-91-37.compute-1.amazonaws.com
The authenticity of host 'ec2-3-87-91-37.compute-1.amazonaws.com (3.87.91.37)'
can't be established.
ED25519 key fingerprint is SHA256:EucMPpJ/1hn1NmVKHXzLMv5Chb2XSRyR2SZHCov3Kwo.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-87-91-37.compute-1.amazonaws.com' (ED25519) to
the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Wed May 27 10:44:53 UTC 2026

System load:  0.37           Temperature:   -273.1 C
Usage of /:   11.0% of 18.25GB Processes:    121
Memory usage: 13%           Users logged in: 0
Swap usage:   0%            IPv4 address for ens5: 172.31.25.147

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

2. Install docker on ec2.

➤ Now after connect with ec2 connected and update

```
ubuntu@ip-172-31-25-147:~$ sudo su -
root@ip-172-31-25-147:~# sudo apt update
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
[136 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
[136 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security InRelease [136 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 Packages
[1483 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main Translation-en
[524 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64 Compone
nts [395 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64 c-n-f M
etadata [32.4 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 P
ackages [16.3 MB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages
[92.6 kB]
```

- Now install docker with command

```
Get:52 http://us-east-1.ec2.archive.ubuntu.com/ubuntu/resolute-backports/multi
rse amd64 c-n-f Metadata [120 B]
Fetched 32.0 MB in 4s (7977 kB/s)
35 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ip-172-31-25-147:~# apt install docker.io -y
Installing:
  docker.io

Installing dependencies:
  bridge-utils  dns-root-data  pigz  ubuntu-fan
  containerd    dnsmasq-base  runc

Suggested packages:
  ifupdown          | cgroup-lite  docker-compose-v2  rootlesskit
  aufs-tools        | debootstrap  docker-doc          | zfs-fuse
  cgroupfs-mount    | docker-buildx  rinse              | zfsutils

Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 35
```

- Now with systemctl command start, enable and check status of docker

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-25-147:~# systemctl start docker
root@ip-172-31-25-147:~# systemctl enable docker
root@ip-172-31-25-147:~# systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: ena
   Active: active (running) since Wed 2026-05-27 10:45:42 UTC; 32s ago
   Invocation: c0d40eb5af44428eb2f5ea9db6c9d568
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 2004 (dockerd)
      Tasks: 8
     Memory: 25.7M (peak: 25.8M)
        CPU: 346ms
     CGroup: /system.slice/docker.service
             └─2004 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/cont
May 27 10:45:42 ip-172-31-25-147 dockerd[2004]: time="2026-05-27T10:45:42.16110
May 27 10:45:42 ip-172-31-25-147 dockerd[2004]: time="2026-05-27T10:45:42.19487
May 27 10:45:42 ip-172-31-25-147 dockerd[2004]: time="2026-05-27T10:45:42.19990
May 27 10:45:42 ip-172-31-25-147 dockerd[2004]: time="2026-05-27T10:45:42.50019
May 27 10:45:42 ip-172-31-25-147 dockerd[2004]: time="2026-05-27T10:45:42.50840
May 27 10:45:42 ip-172-31-25-147 dockerd[2004]: time="2026-05-27T10:45:42.50856
May 27 10:45:42 ip-172-31-25-147 dockerd[2004]: time="2026-05-27T10:45:42.54091
May 27 10:45:42 ip-172-31-25-147 dockerd[2004]: time="2026-05-27T10:45:42.54670
May 27 10:45:42 ip-172-31-25-147 systemd[1]: Started docker.service - Docker Ap
May 27 10:45:42 ip-172-31-25-147 dockerd[2004]: time="2026-05-27T10:45:42.54695
```

- Check docker version

```
root@ip-172-31-25-147:~# docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
root@ip-172-31-25-147:~# sudo usermod -aG docker ubuntu
root@ip-172-31-25-147:~# newgrp docker
```

- Add Ubuntu user to Docker group and apply permissions

```
root@ip-172-31-25-147:~# sudo usermod -aG docker ubuntu
root@ip-172-31-25-147:~# newgrp docker
root@ip-172-31-25-147:~# docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:0e760fd9bc48ba8041e7c6db999bb40bfca508b4be580ac75d32c4e29d202ce1
Status: Downloaded newer image for hello-world:latest
```

3. Pull 5 docker images. (nginx, Apache tomcat, ubuntu, Jenkins, SonarQube)

- Pull nginx docker image

```
root@ip-172-31-25-147:~# docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
5b4d6ff92fc4: Pull complete
5431d0092ffd: Pull complete
7f8b1a2b17d8: Pull complete
b4a248c845e5: Pull complete
45381ecb0e2f: Pull complete
13fd728be9eb: Pull complete
830625e1ac85: Pull complete
cc5f57206478: Download complete
5e6b66b5e5f1: Download complete
Digest: sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

- Pull Apache Tomcat

```
root@ip-172-31-25-147:~# docker pull tomcat
Using default tag: latest
latest: Pulling from library/tomcat
4f4fb700ef54: Pull complete
97c47a5c1a2f: Pull complete
b40150c1c271: Pull complete
b38893372039: Pull complete
65313903a8a8: Pull complete
d92a694d8aa0: Pull complete
6b82a07ff6e7: Pull complete
13883d0c1f97: Download complete
f80ef884f8c2: Download complete
Digest: sha256:247650787b97751d68aa2d532d6e09111c36946d9a50ea0e5071e19facc1e5e7
Status: Downloaded newer image for tomcat:latest
docker.io/library/tomcat:latest
```

- Pull Ubuntu

```
root@ip-172-31-25-147:~# docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
1c24335ddd46: Pull complete
6f5c5aa4e145: Pull complete
9bcf140d7f0f: Download complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
```

- Pull Jenkins

```
Using default tag: latest
latest: Pulling from jenkins/jenkins
dd2c8182e62b: Pull complete
f32f49ce655a: Pull complete
f5a9a47a1ece: Pull complete
ccb49a07ac33: Pull complete
2efd8fae3416: Pull complete
4e3081ec956d: Pull complete
b76936956243: Pull complete
4ebcf05322a6: Pull complete
6db82e8ffefb: Pull complete
938dffaffeaf: Pull complete
f71eb52e5b07: Pull complete
43343e2a5ed9: Pull complete
29d32e0a4d94: Download complete
Digest: sha256:d566c401e8900312ff6cb4019adc5faf19a3ea9c8b93922f6686b822ce691766
Status: Downloaded newer image for jenkins/jenkins:latest
docker.io/jenkins/jenkins:latest
root@ip-172-31-25-147:~# docker pull sonarqube
Using default tag: latest
latest: Pulling from sonarqube/sonarqube
5b4d6ff92fc4: Pull complete
5431d0092ffd: Pull complete
7f8b1a2b17d8: Pull complete
b4a248c845e5: Pull complete
45381ecb0e2f: Pull complete
13fd728be9eb: Pull complete
830625e1ac85: Pull complete
cc5f57206478: Download complete
5e6b66b5e5f1: Download complete
Digest: sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
Status: Downloaded newer image for sonarqube/sonarqube:latest
docker.io/sonarqube/sonarqube:latest
```

- Pull SonarQube

```
docker.io/jenkins/jenkins:latest
root@ip-172-31-25-147:~# docker pull sonarqube
Using default tag: latest
latest: Pulling from library/sonarqube
4f4fb700ef54: Pull complete
c10b8232386e: Pull complete
f4306f81d85a: Pull complete
907ea4a8bc72: Download complete
8506693bc963: Download complete
Digest: sha256:35bedac3f40cda75969890da59b17d577770844fe6ef659206c678a8e00921c7
Status: Downloaded newer image for sonarqube:latest
docker.io/library/sonarqube:latest
```

4. Run nginx container and expose on port 81.

- Verify Downloaded Images

```
docker.io/library/sonarqube:latest
root@ip-172-31-25-147:~# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	0e760dfdbc48	25.9kB	9.49kB	
jenkins/jenkins:latest	d566c401e890	801MB	290MB	
nginx:latest	5aca99593157	241MB	66MB	
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB	
tomcat:latest	247650787b97	589MB	158MB	
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	

```
root@ip-172-31-25-147:~#
```

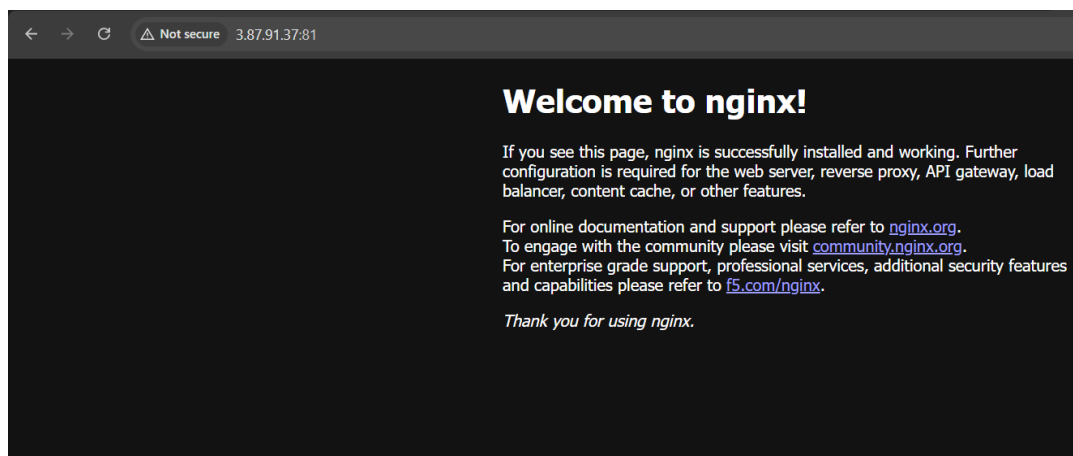
- Run Nginx Container on Port 81
- `docker run -d --name nginx-container -p 81:80 nginx`

```
root@ip-172-31-25-147:~# docker run -d --name nginx-container -p 81:80 nginx
32cb6d249f269c1fb3b0e7798df9005944bca584e4e27448476addec67f7c7a1
root@ip-172-31-25-147:~# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
32cb6d249f26	nginx	"/docker-entrypoint..."	23 seconds ago	Up 22 seconds	0.0.0.0:81->80/tcp, [::]:81->80/tcp	nginx-container

```
root@ip-172-31-25-147:~# docker rmi tomcat
Untagged: tomcat:latest
Deleted: sha256:247650787b97751d68aa2d532d6e09111c36946d9a50ea0e5071e19facc1e5e7
```

- `docker ps` and check with URL in browser



5. Delete the Apache tomcat image from local.

- Delete Apache Tomcat Image from Local
- `docker rmi tomcat`

```
root@ip-172-31-25-147:~# docker rmi tomcat
Untagged: tomcat:latest
Deleted: sha256:247650787b97751d68aa2d532d6e09111c36946d9a50ea0e5071e19facc1e5e7
```

```
root@ip-172-31-25-147:~# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	0e760fdfbc48	25.9kB	9.49kB	U
jenkins/jenkins:latest	d566c401e890	801MB	290MB	
nginx:latest	5aca99593157	241MB	66MB	U
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB	
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	

6. Inspect the Jenkins image, SonarQube image.

- Inspect Jenkins Image
- `docker image inspect jenkins/jenkins`

```
root@ip-172-31-25-147:~# docker image inspect jenkins/jenkins
[
  {
    "Id": "sha256:d566c401e8900312ff6cb4019adc5faf19a3ea9c8b93922f6686b822ce691766",
    "RepoTags": [
      "jenkins/jenkins:latest"
    ],
    "RepoDigests": [
      "jenkins/jenkins@sha256:d566c401e8900312ff6cb4019adc5faf19a3ea9c8b93922f6686b822ce691766"
    ],
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-05-26T07:36:24.873517984Z",
    "Config": {
      "User": "jenkins",
      "ExposedPorts": {
        "50000/tcp": {},
        "8080/tcp": {}
      },
      "Env": [
        "PATH=/opt/java/openjdk/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "LANG=C.UTF-8",
        "JENKINS_HOME=/var/jenkins_home",
        "JENKINS_SLAVE_AGENT_PORT=50000",
        "REF=/usr/share/jenkins/ref",
        "JENKINS_UC=https://updates.jenkins.io",
        "JENKINS_UC_EXPERIMENTAL=https://updates.jenkins.io/experimental",
        "JENKINS_INCREMENTALS_REPO_MIRROR=https://repo.jenkins-ci.org/incrementals",
        "COPY_REFERENCE_FILE_LOG=/var/jenkins_home/copy_reference_file.log",
        "JAVA_HOME=/opt/java/openjdk",
        "JENKINS_VERSION=2.566"
      ]
    }
  ]
]
```

- Inspect SonarQube Image
- `docker image inspect SonarQube`

```

root@ip-172-31-25-147:~# docker image inspect sonarqube
[
  {
    "Id": "sha256:35bedac3f40cda75969890da59b17d57770844fe6ef659206c678a8e00921c7",
    "RepoTags": [
      "sonarqube:latest"
    ],
    "RepoDigests": [
      "sonarqube@sha256:35bedac3f40cda75969890da59b17d57770844fe6ef659206c678a8e00921c7"
    ],
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-05-11T22:13:40.873390969Z",
    "Config": {
      "User": "sonarqube",
      "ExposedPorts": {
        "9000/tcp": {}
      },
      "Env": [
        "PATH=/opt/java/openjdk/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "JAVA_HOME=/opt/java/openjdk",
        "LANG=en_US.UTF-8",
        "LANGUAGE=en_US:en",
        "LC_ALL=en_US.UTF-8",
        "JAVA_VERSION=jdk-25.0.3+9",
        "DOCKER_RUNNING=true",
        "SONARQUBE_HOME=/opt/sonarqube",
        "SONAR_VERSION=26.5.0.122743",
        "SQ_DATA_DIR=/opt/sonarqube/data",
        "SQ_EXTENSIONS_DIR=/opt/sonarqube/extensions",
        "SQ_LOGS_DIR=/opt/sonarqube/logs",
        "SQ_TEMP_DIR=/opt/sonarqube/temp",
        "ES_TMPDIR=/opt/sonarqube/temp"
      ],
      "Entrypoint": [

```

7. Run Jenkins container and run one sample job.

- Run Jenkins:

```

docker run -d \
--name jenkins-container \
-p 8080:8080 \
-p 50000:50000 \
jenkins/jenkins

```

- Check running containers:

```

root@ip-172-31-25-147:~# docker run -d \
--name jenkins-container \
-p 8080:8080 \
-p 50000:50000 \
jenkins/jenkins
3809334d37443ed8ec1ea465448d7bb6c2d9ee67e66e2b984a0ff8298faa18
root@ip-172-31-25-147:~# docker ps
docker: unknown command: docker ps
Run 'docker --help' for more information
root@ip-172-31-25-147:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                                                                                               NAMES
3809334d3744   jenkins/jenkins   "/usr/bin/tini -- /u..."   24 seconds ago   Up 23 seconds   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp   jenkins-container
32cb6d249f26   nginx         "/docker-entrypoint..."   4 minutes ago    Up 4 minutes    0.0.0.0:81->80/tcp, [::]:81->80/tcp                                                                nginx-container

```

- Open Jenkins in browser:

[←](#)
[→](#)
[🔄](#)
Not secure
3.87.91.37:8080/login?from=%2F

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

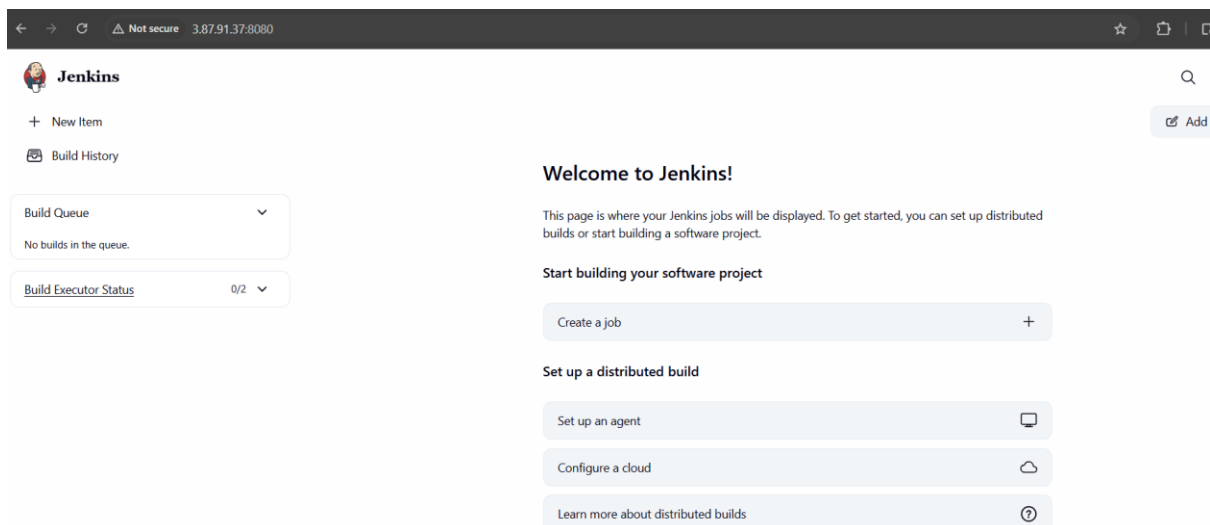
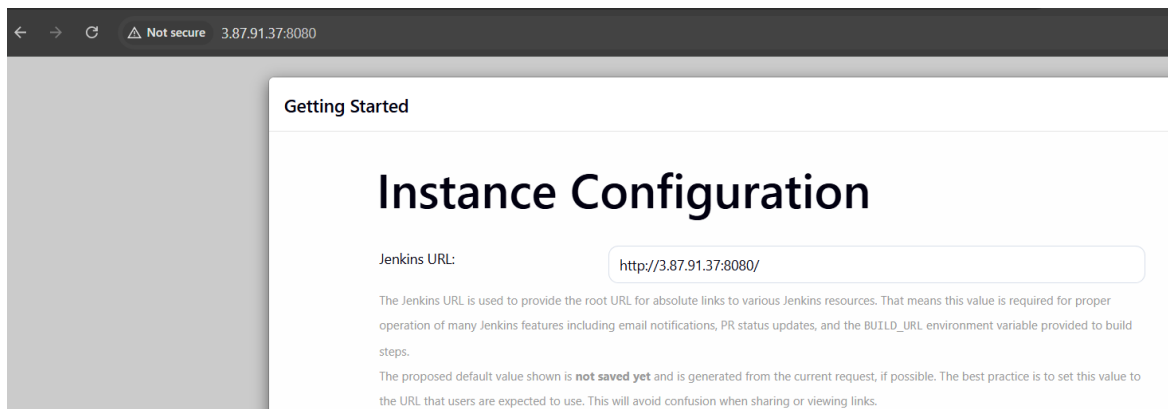
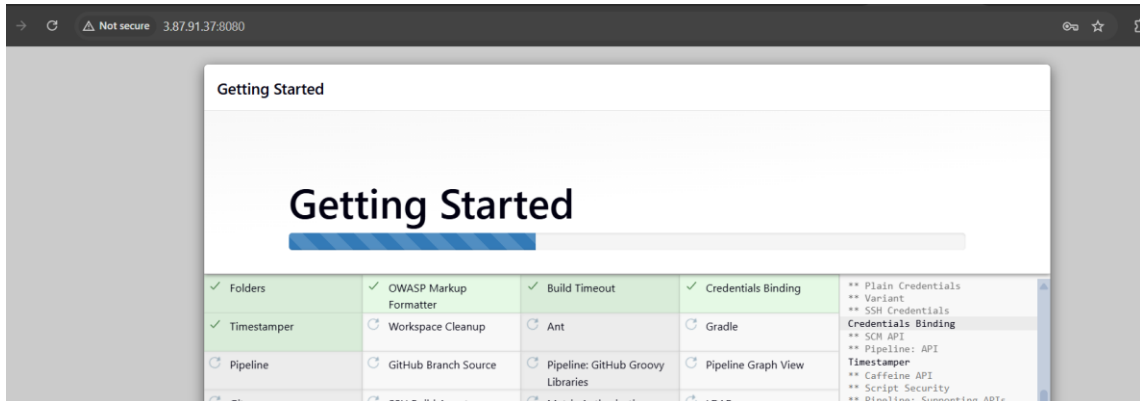
`/var/jenkins_home/secrets/initialAdminPassword`

Please copy the password from either location and paste it below.

Administrator password

➤ Get Jenkins Initial Password

```
32cb6d249f26 nginx "/docker-entrypoint...." 4 minutes ago Up 4 minutes 0.0.0.0:81->80/tcp,
root@ip-172-31-25-147:~# docker exec jenkins-container cat /var/jenkins_home/secrets/initialAdminPassword
c43c2ad8f5914f62a4b26b7c3030a60d
root@ip-172-31-25-147:~#
```



- Create Sample Jenkins Job
- Create Freestyle Job

Jenkins / All / New Item

New Item

Enter an item name

sample-job

Select an item type

- Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
- Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Multi-configuration project

➤ Add Build Step

Jenkins / sample-job / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

☐ Add timestamps to the Console Output

☐ Inspect build log for published build scans

☐ Terminate a build if it's stuck

☐ With Ant ?

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

- Execute shell ?

Command

See [the list of available environment variables](#)

```
echo "Hello from Jenkins Docker Container"
```

➤ Build the Job and check console output

Jenkins / sample-job / #1

Console

- Status
- Changes
- Console Output
- Delete build #1
- Timings
- Edit build information

```
Started by user Kathi Arunsai
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/sample-job
[sample-job] $ /bin/sh -xe /tmp/jenkins10115389038515766646.sh
+ echo Hello from Jenkins Docker Container
Hello from Jenkins Docker Container
+ date
Wed May 27 11:02:05 UTC 2026
Finished: SUCCESS
```